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DisentangledTMR: Privacy-Preserving Skeleton Motion Retargeting via Factorized Transformers

Keep **what** the body is doing. Replace **who** is doing it.

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PRIVACY GAIN
75.4 → 12.2%
Re-identification falls from **30× chance** to **4.9×**, against a 2.5% floor

UTILITY KEPT
55.4%
Action recognition on retargeted skeletons: **+12.3 pts over DMR, +35.5 over PMR**

PARETO
Both axes
The only method that beats **both** DMR and PMR on privacy *and* utility

TUNABLE
β = 0.2
One knob trades privacy for utility at test time: **76.8% AR at 17.3% RI**

The problem

3D skeletons look anonymous, but bone lengths, limb proportions and gait are a **biometric signature**: an off-the-shelf recogniser re-identifies the actor in raw NTU skeletons **75.4% of the time** across 40 identities, 30× chance.

Yet skeletons are exactly the modality we want to share for health monitoring, elder care and behaviour analysis. Blurring or adding noise destroys the action along with the identity.

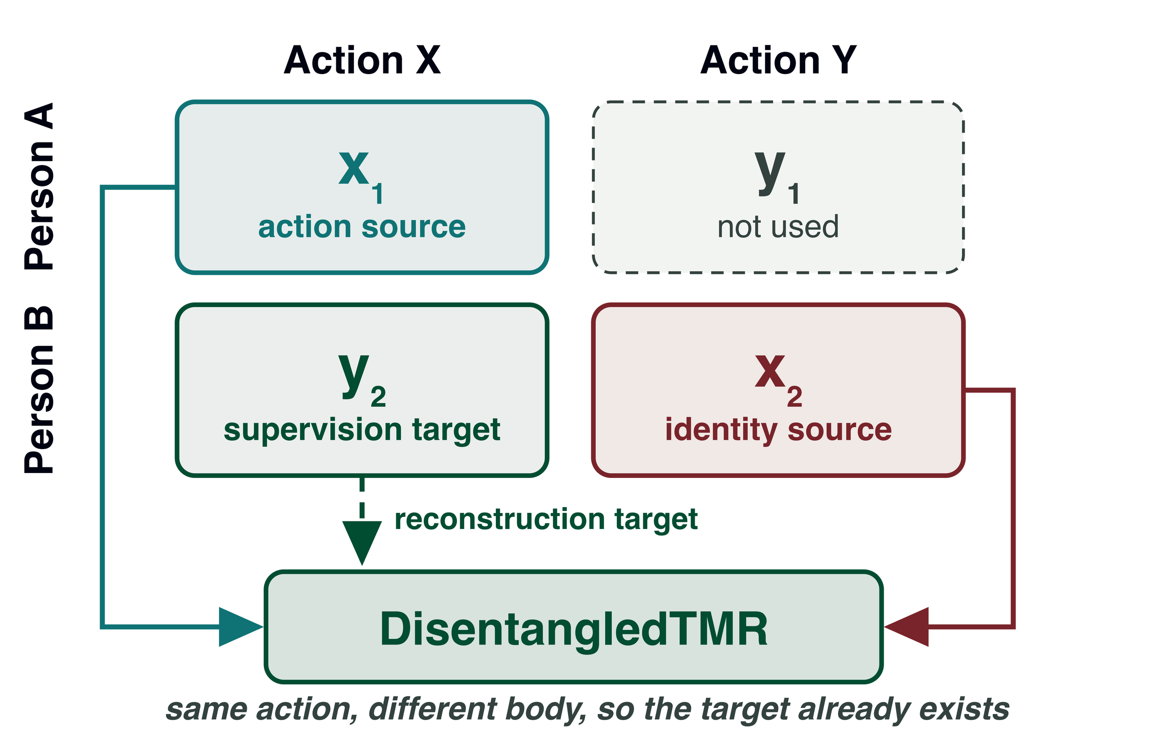
The idea

Replace identity rather than removing it. Encode the motion of one person and the body of another into two separate latent streams, then decode a sequence that performs the **source action** on the **target skeleton**.

- **Action stream** (768-d) captures dynamics: position, velocity, acceleration → multi-scale temporal convolution → attention.
- **Identity stream** (256-d) captures structure: mean static pose + bone lengths → spatial GCN. Deliberately **low-capacity**, an information bottleneck that starves it of motion detail.
- **Asymmetric by design.** The architecture does most of the disentangling; the contrastive, adversarial, orthogonality and mutual-information losses do the rest.

Training signal

Every training sample is a **cross-identity quadruplet**: two actors who each perform the same two actions, so the exact retargeting target already exists in the data and the reconstruction term needs no adversarial proxy.



Three-stage training

Order is mandatory. Skipping a stage measurably degrades disentanglement.

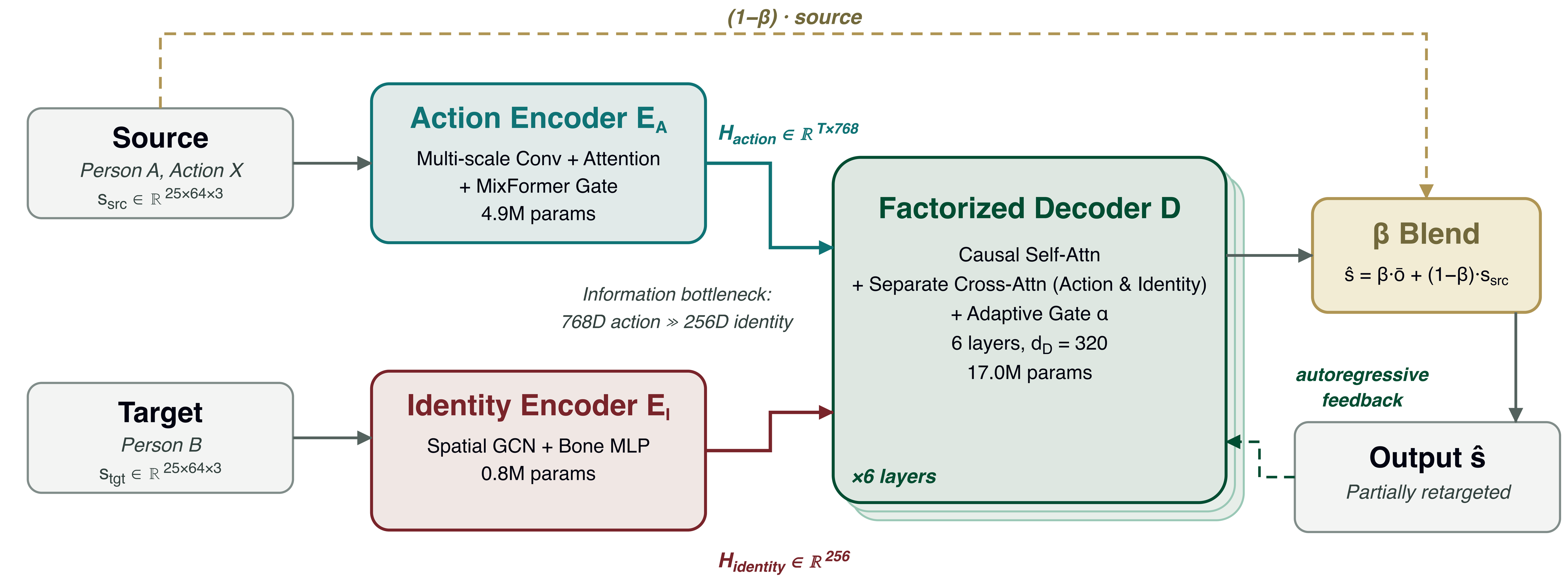
1. **Encoder pre-training.** Both encoders learn with classification heads plus the four disentanglement losses. Decoder frozen.
2. **Decoder training.** Reconstruction plus physical plausibility (bone length, smoothness, velocity, end-effector, foot contact, joint limits). Teacher forcing 1.0 → 0.5.
3. **End-to-end fine-tuning.** All components jointly, teacher forcing 0.5 → 0.3, with output-level adversarial and cooperative heads.

Scope

Data. NTU RGB+D 60 / 120 and ETRI-Activity3D, single-person sequences only (two-person NTU actions excluded).

Privacy is measured against trained recognisers, not a worst-case adversary with full knowledge of the retargeting model.

Architecture

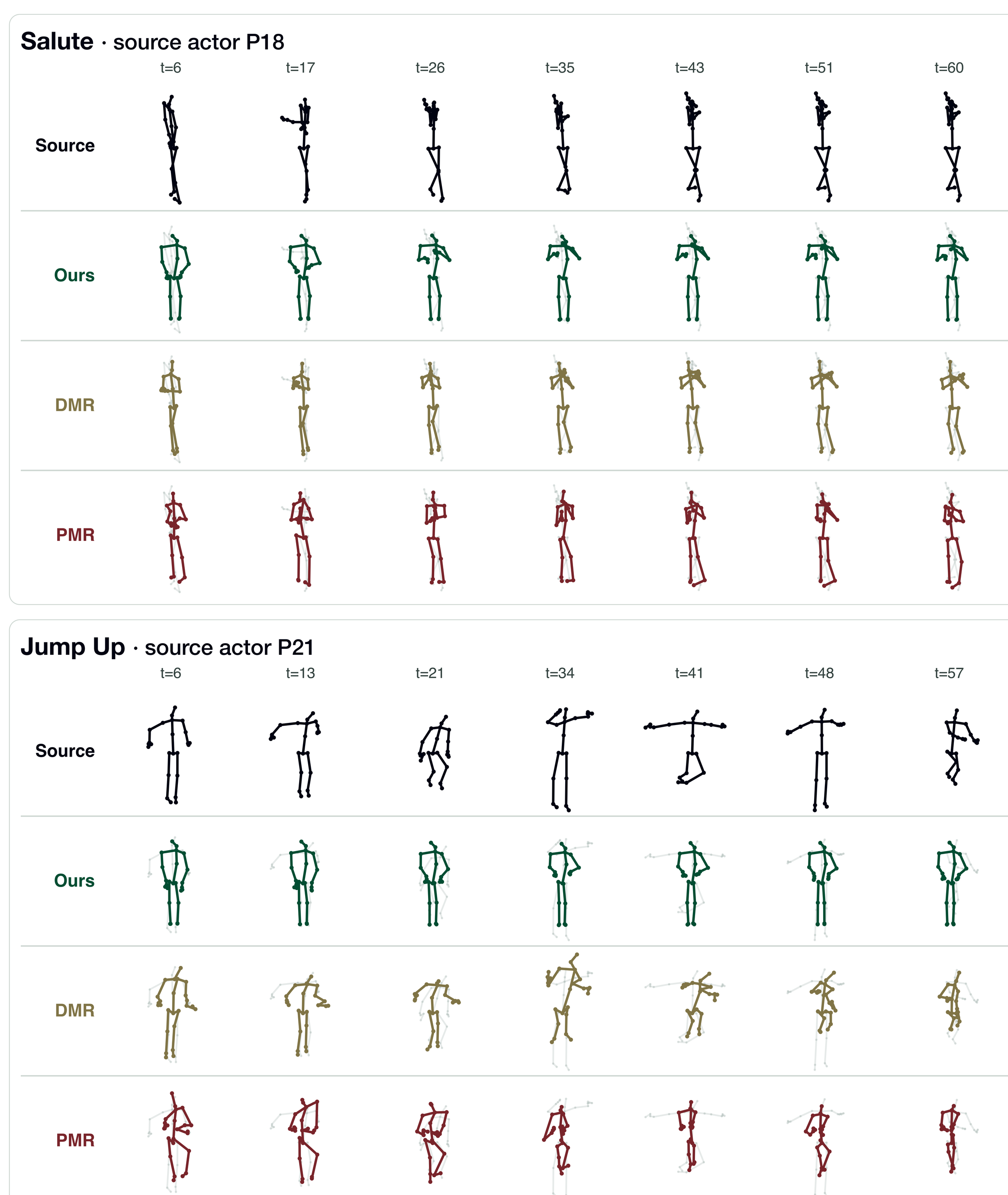


The decoder attends to each stream separately, then blends them with a learned per-channel gate **α**. Generation is autoregressive under a causal mask.

22.5M parameters, three quarters of them in the decoder, against 4.9M for DMR and 1.0M for PMR.

Retargeting runs once, offline, at collection time. Downstream consumers of the shared skeletons pay nothing.

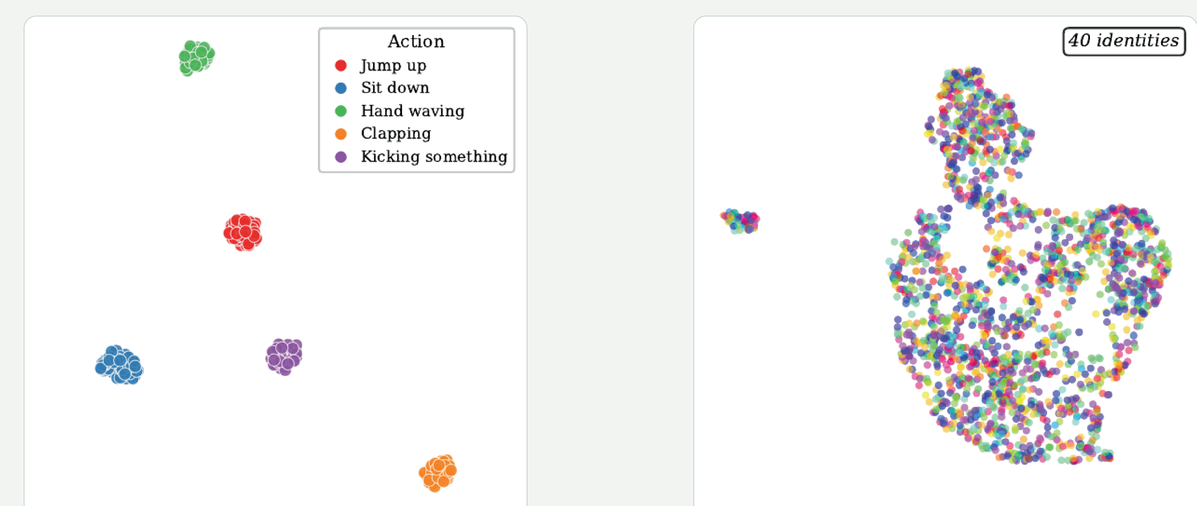
Qualitative results



Real sequences from NTU RGB+D 60. **Grey ghost** = source motion; coloured overlay = each method's output. Ours preserves the action on a new body; **DMR** drifts off-posture and **PMR** collapses the skeleton, consistent with its 19.9% action accuracy.

Is it disentangled?

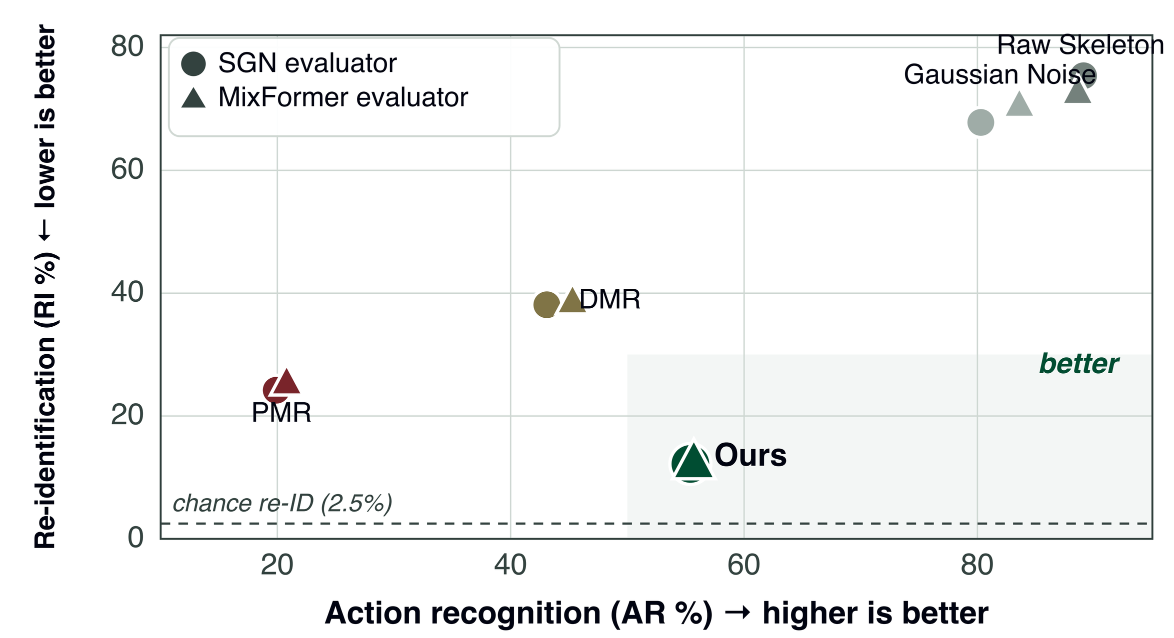
UMAP of the **action embedding**, taken from the action-recognition classifier's features.



Five action classes
Each class lands in its own tight cluster. The action code carries the semantics.

40 identities
Colouring the action embedding by actor leaves no recoverable structure.

Privacy–utility landscape



Down and to the right is better. DMR trades little privacy for moderate utility; PMR destroys both. DisentangledTMR sits alone in the favourable quadrant.

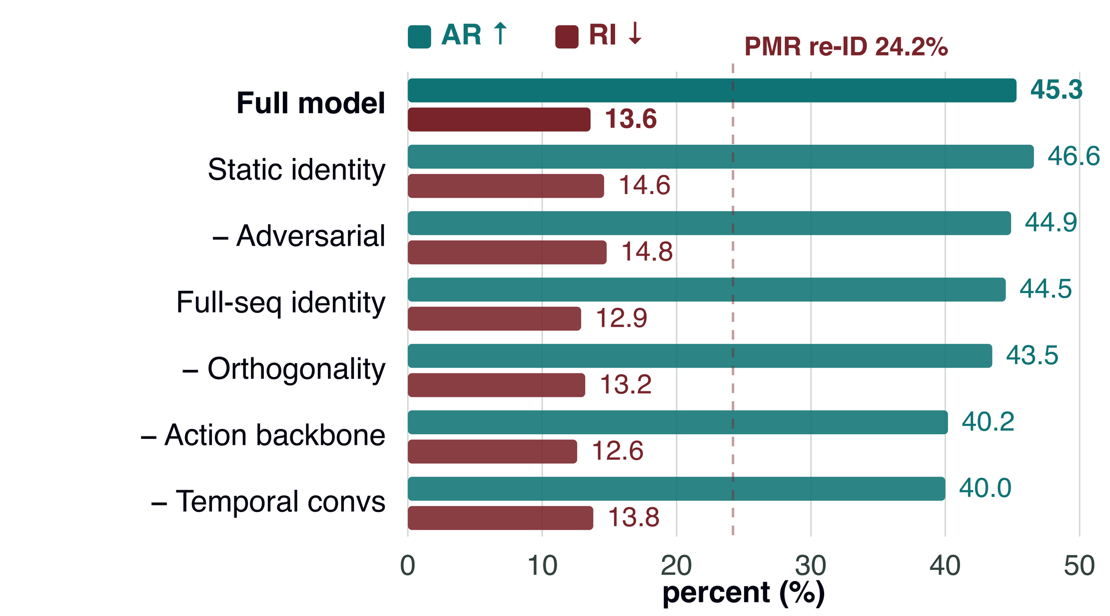
Main results

NTU RGB+D 60, cross-view. AR = action recognition (higher better), RI = re-identification of the *source* actor (lower better). Chance RI = 2.5%.

	SGN		MIXFORMER	
METHOD	AR ↑	RI ↓	AR ↑	RI ↓
Raw skeleton	89.1	75.4	88.6	72.8
Gaussian noise	80.3	67.8	83.6	70.7
DMR	43.1	38.1	45.3	38.7
PMR	19.9	24.2	20.8	25.5
DisentangledTMR	**55.4**	**12.2**	**55.7**	**12.5**

Mean of 3 seeds; ±0.7 AR / ±0.2 RI (SGN). Ours is the only method more private than PMR *and* more useful than DMR, under both evaluators.

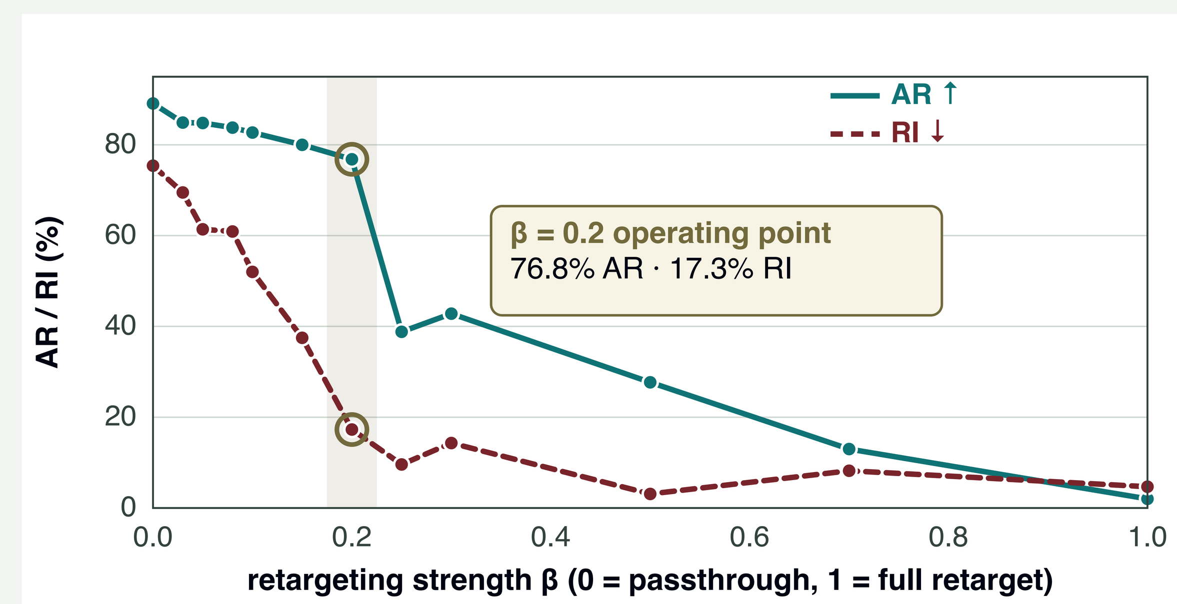
Ablations



Removing the **action backbone** or the **temporal convolutions** costs ~5 points of accuracy: utility lives in the action encoder's capacity. **Every variant still beats PMR's 24.2% RI**, so the factorised architecture carries the privacy gain, not any single loss term.

The sweep runs at batch size 32, so compare rows here rather than against the main table. Rows prefixed – drop that component; the two identity rows change how the identity stream is pooled.

One knob at test time



Frozen recognisers score every β, so the operating point sweeps *without retraining*. At **β = 0.2** the curve knees: re-identification has already fallen 75.4 → 17.3% while action accuracy still holds at 76.8%. Deployments pick β from their own risk budget.

Takeaways

- Skeleton data is **not** anonymous; treat it as biometric.
- **Replacing** identity beats suppressing it: noise loses utility faster than it buys privacy.
- **Asymmetric architecture** does the disentangling; the losses only refine it.
- A single post-hoc **β** exposes the whole privacy–utility curve from one trained model.